EXTENDED EUCLIDS WITH BEZOUTS COEFFICIENT

Problem Submissions Leaderboard

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Euclid's Identity states that for any two positive integers A and B, there exist integers x and y such that: Ax + By = D

where D is the Greatest Common Divisor (GCD) of A and B. Given two positive integers A and B, find: x y D = gcd(A, B)

such that:

Ax + By = D

If multiple pairs (x, y) satisfy the equation, choose the pair with the minimum value of: |x| + |y|

If there is still more than one pair, choose the pair for which:

x <= y

Input Format

A B

Constraints

1 ≤ A, B ≤ 10^9

Output Format

x y D D is gcd(A, B) Ax + By = D (x, y) satisfies the minimum-selection rules.

Sample Input 0

```
4 6
```

Sample Output 0

```
-1 1 2
```

Change Theme Language Python 3

```
# Enter your code here. Read input from STDIN. Print output to STDOUT
A,B=list(map(int,input().split()))
a,b=A,B
s0,s1=1,0
t0,t1=0,1
while b!=0:
    q=a//b
    a,b=b,a-q*b
    s0,s1=s1,s0-q*s1
    t0,t1=t1,t0-q*t1
print(s0,t0,a)

```

Line: 12 Col: 1

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